

Experiment No. 19

Aim: To prepare pure crystals of copper sulfate ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$) or benzoic acid $\text{C}_6\text{H}_5\text{-COOH}$ from a given impure sample.

Theory: Generally the substance is more soluble at higher temperature in a given solvent.

In chemistry, recrystallization is a technique used to purify chemicals. By dissolving both impurities and a compound in an appropriate solvent, either the desired compound or impurities can be removed from the solution, leaving the other behind with filter paper.

Apparatus: Glass beakers, glass rod, funnel, filter paper, evaporating dish, wire gauze, tripod stand, Bunsen burner, etc.

Procedure:

1. 5 gm of an impure sample of copper sulfate or benzoic acid is dissolved at high temperature in minimum quantity of water.
2. The solution is heated to get the saturated solution. This hot solution is filtered to remove insoluble impurities, evaporate this solution in evaporating dish.
3. The filtrate is allowed to cool slowly to room temperature when fine crystals of pure substance are formed.
4. These crystals are removed from the mother liquor by filtration and washed with the same solvent to remove adhered impurities. These crystals are then dried.

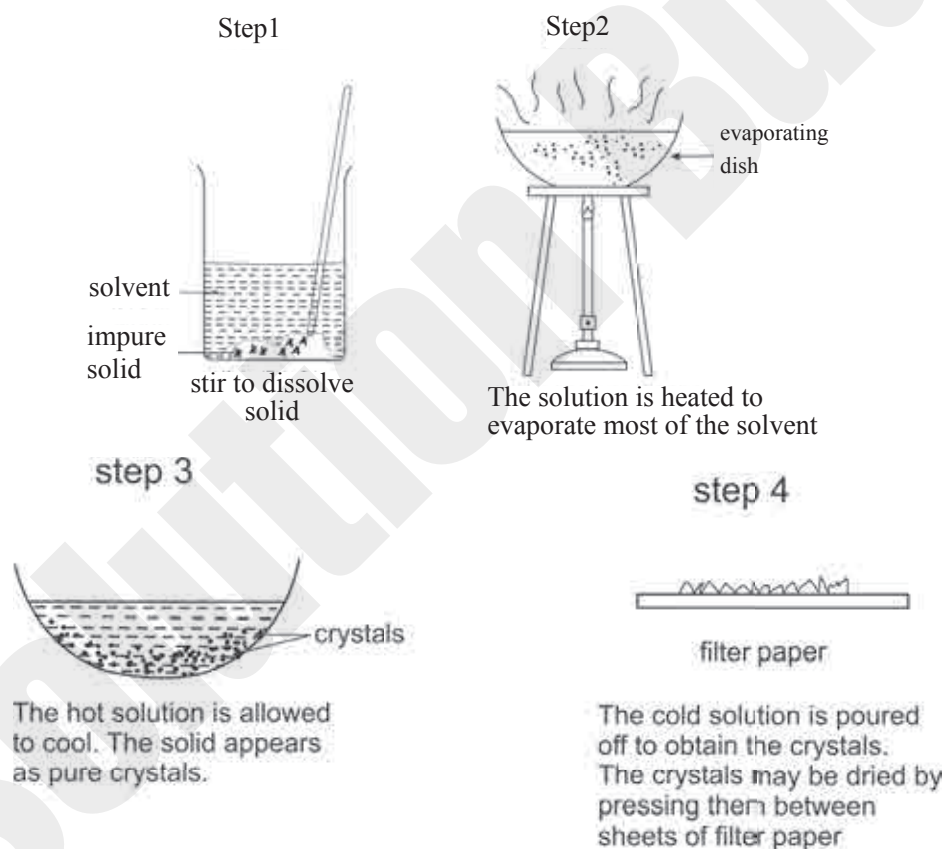


Fig. 19.1 Simple crystallization process.

Remark and sign of teacher:

MCQ

Select [✓] or underline the most appropriate answer from given alternatives of each sub question.

- At which temperature generally the substance is more soluble in a given solvent?
a. Very very low c. Very low
b. At room temperature d. Higher temperature.
- The stage at which crystal starts to appear on glass rod is called -----.
a. boiling point c. melting point
b. point of crystallization d. triple point
- Which solution is to be prepared for crystallization?
a. Unsaturated c. Supersaturated
b. Saturated d. Fused mass of impure substance
- What is the Color of prismatic copper sulfate crystals?
a. Red b. Blue c. Pink d. Green
- What is the Color of crystals of benzoic acid?
a. Black b. Yellow c. White d. Brown

Short answer questions

1. How does cooling rate affect crystal size?

Ans. Slow cooling produces large and well-defined crystals, whereas rapid cooling produces small crystals.

2. Which type of solvent is used for crystallization?

Ans. A suitable solvent in which the substance is highly soluble at high temperature and sparingly soluble at low temperature is used for crystallization.

3. Write the names of observed Color of saturated solutions of copper sulfate and benzoic acid in water at high temperature.

Ans. The saturated solution of copper sulfate is blue, while the saturated solution of benzoic acid is colorless.

4. Write the names of apparatus used during filtration process.

Ans. The apparatus used for filtration are filter funnel, filter paper, beaker, conical flask and glass rod.

5. What is filtrate?

Ans. The liquid that passes through the filter paper during filtration is called the filtrate.

6. Arrange the following steps involved in crystallization in proper sequence.

a. Filtration b. Dissolution c. Drying d. Crystallization e. Solvent selection.
Ans. a. Filtration b. Dissolution
c. Drying d. Crystallization
e. Solvent selection

Remark and sign of teacher: